

Final Mock Exam — Set A — Solutions

Lectures 1–12 (comprehensive)

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Summer Semester 2026

Overview of the final mock exam — Set A

Problem	Topic	Lecture slides	Points
P1	Moving beyond linearity — splines (Ch. 7)	Moving Beyond Linearity (Lec. 9)	16
P2	Generalised additive models (Ch. 7)	Moving Beyond Linearity (Lec. 9)	8
P3	Regression & classification trees (Ch. 8)	Tree-Based Methods (Lec. 10)	20
P4	Ensembles: bagging/RF/boosting (Ch. 8)	Tree-Based Methods (Lec. 10)	16
P5	Neural networks by hand (Ch. 10)	Deep Learning (Lec. 11)	20
P6	Multiple testing (Ch. 13)	Multiple Testing (Lec. 12)	20
P7	Cumulative essentials (Ch. 2–6)	Lectures 1–8	20
Σ			120

Exam conditions

120 minutes for 120 points (≈ 1 point per minute). Allowed aids: non-programmable calculator and one handwritten A4 sheet (both sides). All required constants (e-powers and logs) are given in the problems. Justify every answer.

Problem 1 — task

Problem 1 (16 P) — Moving beyond linearity

- (a) [4 P] Degrees of freedom of a cubic spline with K interior knots — justify by counting parameters and constraints; evaluate for $K = 5$.
- (b) [4 P] Degrees of freedom of a *natural* cubic spline with K knots ($K = 5$); which extra constraint acts where and why is it useful?
- (c) [4 P] Smoothing spline $\sum_i (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$: role of the two terms; behaviour of $\hat{g}$ as $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$.
- (d) [4 P] Truncated-power basis of a cubic spline with one knot ξ ; define $(x - \xi)_+^3$; why does adding it keep the fit smooth at ξ ?

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

Problem 1 — solution (a) and (b)

Solution P1 — (a) and (b)

(a) K knots $\Rightarrow K + 1$ cubic pieces $\times 4$ coefficients $= 4K + 4$ parameters; continuity of g and g' and g'' at each knot $\Rightarrow 3K$ constraints:

$$\text{df} = 4(K + 1) - 3K = K + 4 \quad \Rightarrow \quad K = 5 : \text{df} = 9.$$

(Equivalently: basis $1, x, x^2, x^3$ plus one truncated power per knot $= 4 + K$.)

(b) Natural spline: *linear beyond the boundary knots* (regions $X < \xi_1$ and $X > \xi_K$); killing the quadratic and cubic terms there costs 2 parameters per boundary region:

$$\text{df} = (K + 4) - 4 = K \quad \Rightarrow \quad K = 5 : \text{df} = 5.$$

Why useful: splines are most erratic near the boundaries where data are sparse; the linearity constraint stabilises the fit there (lower variance for a little extra bias).

Problem 1 — solution (c) and (d)

Solution P1 — (c) and (d)

(c) RSS term = fidelity to the data; $\lambda \int g''(t)^2 dt$ penalises curvature (wiggleness); λ trades the two off.

- $\lambda \rightarrow 0$: no penalty $\Rightarrow \hat{g}$ *interpolates* the training points exactly — extremely wiggly, zero training RSS, severe overfitting (effective df $\rightarrow n$).
- $\lambda \rightarrow \infty$: curvature infinitely expensive $\Rightarrow g'' \equiv 0 \Rightarrow \hat{g}$ is the *least-squares straight line* (effective df $\rightarrow 2$).

λ (chosen by LOOCV) steers the effective df smoothly between n and 2.

(d) $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3$ with

$$(x - \xi)_+^3 = \begin{cases} (x - \xi)^3 & x > \xi \\ 0 & x \leq \xi \end{cases}$$

$(x - \xi)_+^3$ has value and first and second derivative 0 at $\xi \Rightarrow$ only the third derivative jumps: the fit stays continuous with continuous g' and g'' — a cubic spline.

Problem 2 — task

Problem 2 (8 P) — Generalised additive models

- (a) [3 P] General form of a GAM for quantitative Y ; how does it extend multiple linear regression?
- (b) [3 P] One advantage concerning the interpretation of the f_j ; the main structural limitation — and its repair.
- (c) [2 P] Backfitting in one or two sentences.

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

Problem 2 — solution

Solution P2 — GAMs

(a) $Y = \beta_0 + f_1(X_1) + f_2(X_2) + \dots + f_p(X_p) + \varepsilon$.

Linear regression is the special case $f_j(X_j) = \beta_j X_j$; a GAM replaces each linear term by a smooth non-linear f_j (spline, local regression) but keeps the *additive* structure so the effects still simply add up.

(b) *Advantage*: additivity $\Rightarrow$ the effect of each X_j is its own function f_j , which can be plotted and interpreted individually while the other predictors are held fixed — one panel per predictor, as readable as a coefficient table.

Limitation: no *interactions* — the effect of X_j cannot depend on X_k .

Repair: add interaction components manually, e.g. $X_j \times X_k$ or a two-dimensional smooth $f_{jk}(X_j, X_k)$.

(c) *Backfitting*: cycle through $j = 1, \dots, p$ and re-fit each f_j to the partial residuals $r_i = y_i - \beta_0 - \sum_{k \neq j} f_k(x_{ik})$ with all other functions held fixed; repeat until convergence.

Problem 3 — task

Problem 3 (20 P) — Trees by hand

Number of promotional events x and monthly revenue y (1,000 EUR) for $n = 6$ café branches; one split at s : $R_1 = \{x < s\}$ and $R_2 = \{x \geq s\}$ with region-mean predictions.

i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	4	6	8	14	16	18

(a) **[8 P]** Evaluate the cutpoints $s = 2.5$ and $s = 3.5$ by total RSS; which is chosen? (b) **[2 P]** Prediction for a new branch with $x = 5$. (c) **[6 P]** Node with 7 “Stay” / 3 “Leave”: proportions; Gini; entropy ($\ln 0.7 = -0.3567$; $\ln 0.3 = -1.2040$); predicted class. (d) **[4 P]** Cost-complexity pruning: role of α in $\sum_m \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$.

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

Problem 3 — solution (a) and (b)

Solution P3 — (a) and (b)

(a) Root mean $\bar{y} = 66/6 = 11$. Region means and RSS per cutpoint:

s	R_1 (y-values)	$\hat{y}_{R_1}$	R_2 (y-values)	$\hat{y}_{R_2}$	total RSS
2.5	{4,6}	5	{8,14,16,18}	14	$2 + 56 = 58$
3.5	{4,6,8}	6	{14,16,18}	16	$8 + 8 = 16$

Details: $s = 2.5$: $(4-5)^2 + (6-5)^2 = 2$; $(8-14)^2 + (14-14)^2 + (16-14)^2 + (18-14)^2 = 36 + 0 + 4 + 16 = 56$.

$s = 3.5$: $(4-6)^2 + 0 + (8-6)^2 = 8$; $(14-16)^2 + 0 + (18-16)^2 = 8$.

Greedy recursive binary splitting picks the smaller RSS: $s = 3.5$ ($16 \ll 58$; root RSS was 166) — it cleanly separates the three low-revenue from the three high-revenue branches.

(b) $x = 5 \geq 3.5 \Rightarrow$ region $R_2 \Rightarrow \hat{y} = \hat{y}_{R_2} = 16$ (i.e. 16,000 EUR per month).

Problem 3 — solution (c) and (d)

Solution P3 — (c) and (d)

(c) $\hat{p}_{\text{Stay}} = 7/10 = 0.7$; $\hat{p}_{\text{Leave}} = 3/10 = 0.3$.

$$G = 0.7 \cdot 0.3 + 0.3 \cdot 0.7 = 0.21 + 0.21 = 0.42$$

$$D = -(0.7 \ln 0.7 + 0.3 \ln 0.3) = 0.7 \cdot 0.3567 + 0.3 \cdot 1.2040 = 0.2497 + 0.3612 = 0.611$$

Predicted class: majority = **Stay** (estimated probability 0.7). Both measures are 0 for pure nodes and largest at a 50:50 split — this node is moderately impure.

(d) α = price per terminal node $|T|$:

- $\alpha = 0$: criterion = training RSS $\Rightarrow$ full tree T_0 is optimal.
- $\alpha \uparrow$: leaves must pay for themselves; branches with small RSS gain are pruned $\Rightarrow$ nested sequence of ever smaller subtrees (weakest-link pruning).
- Choice: *cross-validation* over α ; refit the chosen subtree on all data — balances bias against variance.

Problem 4 — task

Problem 4 (16 P) — Ensembles

- (a) [6 P] B identically distributed trees with variance σ^2 and pairwise correlation ρ :
 $\text{Var}\left(\frac{1}{B} \sum_b \hat{f}_b(x)\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. (i) Why does this show variance reduction and what limits it? (ii) Evaluate for $B = 200$ and $\rho = 0.5$ and $\sigma^2 = 9$; compare with one tree. (iii) Limit as $B \rightarrow \infty$?
- (b) [4 P] Random forests: modification at each split; recommended m for *regression* with $p = 15$; why does it improve on bagging (use the formula)?
- (c) [4 P] Boosting: the three tuning parameters; the slow-learning idea.
- (d) [2 P] What does the OOB error estimate and why is it free?

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

Problem 4 — solution (a) and (b)

Solution P4 — (a) and (b)

(a) (i) Averaging drives the second term $\frac{1-\rho}{B}\sigma^2 \rightarrow 0$ (the independent part of the trees' variability); the first term $\rho\sigma^2$ is independent of B — positive correlation between bootstrap trees limits the reduction.

(ii) $\text{Var} = 0.5 \cdot 9 + \frac{1-0.5}{200} \cdot 9 = 4.5 + 0.0225 = 4.5225$ vs. $\sigma^2 = 9$ for a single tree ($\approx 50\%$ less; exactly 49.75%).

(iii) $B \rightarrow \infty$: variance $\rightarrow \rho\sigma^2 = 4.5$ (floor; more trees never overfit, they only stabilise).

(b) At *each split* only a random subset of m of the p predictors may be used; regression default $m \approx p/3 \Rightarrow p = 15$: $m = 15/3 = 5$ (classification uses $m \approx \sqrt{p}$).

With bagging a dominant predictor tops nearly every tree $\Rightarrow$ trees alike $\Rightarrow \rho$ high. Restricting candidates *decorrelates* the trees ($\rho \downarrow$), which lowers the floor $\rho\sigma^2$ — the part of the variance that averaging alone cannot remove.

Problem 4 — solution (c) and (d)

Solution P4 — (c) and (d)

(c) Boosting for regression trees:

- B — **number of trees**: added sequentially; too many $\Rightarrow$ slow overfitting; choose by CV.
- λ — **shrinkage** (e.g. 0.01 or 0.001): scales each new tree's contribution; smaller $\lambda \Rightarrow$ slower learning $\Rightarrow$ larger B needed.
- d — **splits per tree** (interaction depth): complexity and interaction order of each tree; often stumps ($d = 1$) suffice (additive model).

Slow learning: each tree is fitted to the current *residuals* and only the fraction λ is added — the model improves gradually exactly where it still errs, instead of fitting the data hard at once; this tends to generalise better.

(d) Each bootstrap sample misses $\approx 1/3$ of the observations (out-of-bag); predicting every observation only with trees that never saw it yields the **OOB error** — a valid *test-error* estimate, free as a by-product of the bootstrap (no validation set or CV loop needed).

Problem 5 — task

Problem 5 (20 P) — Neural networks by hand

Input $x = (2,3)$; hidden ReLU units $g(z) = \max\{0,z\}$; linear output:

$$A_1 = g(1 + 2x_1 - 1x_2) \quad A_2 = g(-2 - 1x_1 + 1x_2) \quad f(x) = -1 + 3A_1 + 2A_2$$

- (a) [8 P] Forward pass step by step: pre-activations; A_1 and A_2 ; output $f(x)$. What did the ReLU do to unit 2 and why are non-linear activations essential?
- (b) [6 P] Logits $z = (2.0, 1.0, -1.0)$; true class 1: all three softmax probabilities and the cross-entropy loss ($e^2 = 7.389$; $e^1 = 2.718$; $e^{-1} = 0.368$; $\ln 10.475 = 2.349$).
- (c) [4 P] Parameter count of a $p = 5 \rightarrow k = 4 \rightarrow K = 3$ network with biases (show both layers).
- (d) [2 P] Conv layer: 28×28 input; $F = 3$; $P = 1$; $S = 1$: output size via $(W - F + 2P)/S + 1$.

Lecture slides: Chapter 10 — Deep Learning (Lecture 11).

Problem 5 — solution (a) forward pass

Solution P5 — (a) forward pass

Pre-activations:

$$z_1 = 1 + 2 \cdot 2 - 1 \cdot 3 = 2$$

$$z_2 = -2 - 1 \cdot 2 + 1 \cdot 3 = -1$$

ReLU activations:

$$A_1 = \max\{0, 2\} = 2$$

$$A_2 = \max\{0, -1\} = 0$$

Output:

$$f(x) = -1 + 3 \cdot 2 + 2 \cdot 0 = 5$$

The ReLU *switched unit 2 off*: its negative pre-activation is clipped to 0, so it contributes nothing for this input (though it can for others).

Why essential: with a linear g the whole network collapses into one linear function of x regardless of depth — the non-linearity is what lets hidden layers build flexible functions (piecewise-linear surfaces with ReLU).

Problem 5 — solution (b) softmax and cross-entropy

Solution P5 — (b) softmax and cross-entropy

Exponentials and their sum:

$$e^{2.0} = 7.389 \quad e^{1.0} = 2.718 \quad e^{-1.0} = 0.368 \quad \Rightarrow \quad 7.389 + 2.718 + 0.368 = 10.475$$

Softmax probabilities:

$$\hat{p}_1 = \frac{7.389}{10.475} = 0.705 \quad \hat{p}_2 = \frac{2.718}{10.475} = 0.259 \quad \hat{p}_3 = \frac{0.368}{10.475} = 0.035$$

(sum = 1.000 ✓). Cross-entropy for true class 1:

$$-\ln \hat{p}_1 = -\ln \frac{e^2}{10.475} = -(2 - \ln 10.475) = 2.349 - 2 = 0.349$$

Sanity checks: a perfect prediction ($\hat{p}_1 = 1$) gives loss 0; predicting the uniform $1/3$ would give $\ln 3 \approx 1.099$ — here the network is right and fairly confident.

Problem 5 — solution (c) and (d)

Solution P5 — (c) parameters and (d) convolution

(c) Fully connected $5 \rightarrow 4 \rightarrow 3$ with biases:

Layer	Weights	Biases	Parameters
input $\rightarrow$ hidden	$5 \cdot 4 = 20$	4	24
hidden $\rightarrow$ output	$4 \cdot 3 = 12$	3	15
total			39

(d) Output width = $\frac{W - F + 2P}{S} + 1 = \frac{28 - 3 + 2 \cdot 1}{1} + 1 = 27 + 1 = 28 \Rightarrow$ output feature map 28×28 :
with $F = 3$ and $P = 1$ and $S = 1$ this is the “same” convolution — spatial size preserved (one map per filter).

Problem 6 — task

Problem 6 (20 P) — Multiple testing

- (a) **[4 P]** $m = 25$ independent tests at $\alpha = 0.05$; all nulls true: define and compute the FWER $((0.95)^{25} = 0.2774)$; interpret.
- (b) **[7 P]** Six ordered p -values 0.002; 0.007; 0.009; 0.020; 0.040; 0.320 — FWER control at $\alpha = 0.05$ with (i) Bonferroni and (ii) Holm; thresholds; rejected hypotheses; why is Holm never weaker than Bonferroni?
- (c) **[6 P]** Benjamini–Hochberg at $q = 0.10$ on the same p -values: thresholds jq/m ; rejections; what exactly does FDR control guarantee?
- (d) **[3 P]** Confirmatory phase-III trial with $m = 4$ pre-specified primary endpoints supporting drug approval: FWER or FDR? Justify.

Lecture slides: Chapter 13 — Multiple Testing (Lecture 12).

Problem 6 — solution (a) FWER

Solution P6 — (a) FWER for 25 independent tests

$\text{FWER} = \Pr(V \geq 1)$ — the probability of *at least one* false rejection among the m tests (V = number of falsely rejected true nulls).

With all 25 nulls true and independent tests at level 0.05:

$$\text{FWER} = 1 - \Pr(\text{no false rejection}) = 1 - (1 - 0.05)^{25} = 1 - (0.95)^{25} = 1 - 0.2774 = 0.7226$$

Interpretation: although each test individually errs with probability 5% only, the chance of at least one spurious “discovery” among 25 unadjusted tests is about 72% — multiple testing without adjustment almost guarantees false findings (with $m = 50$: $1 - 0.95^{50} \approx 0.923$).

Problem 6 — solution (b) Bonferroni and Holm

Solution P6 — (b) Bonferroni and Holm at level 0.05

Bonferroni: reject iff $p_{(j)} \leq \alpha/m = 0.05/6 = 0.00833$: only 0.002 and 0.007 pass ($0.009 > 0.00833$) $\Rightarrow$ reject $H_{(1)}$ and $H_{(2)}$: **2 rejections**.

Holm (step-down): compare $p_{(j)}$ with $\alpha/(m - j + 1)$; stop at first failure:

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.007	0.009	0.020	0.040	0.320
$\alpha/(m - j + 1)$	0.00833	0.0100	0.0125	0.0167	0.0250	0.0500
decision	✓	✓	✓	stop	—	—

$0.009 \leq 0.0125$ but $0.020 > 0.0167 \Rightarrow$ reject $H_{(1)}$ and $H_{(2)}$ and $H_{(3)}$: **3 rejections**.

Holm $\supseteq$ Bonferroni: Holm's hurdles $\alpha/(m - j + 1) \geq \alpha/m$ for all j (equality only at $j = 1$) $\Rightarrow$ uniformly at least as powerful — and it still controls the FWER without independence assumptions.

Problem 6 — solution (c) and (d)

Solution P6 — (c) Benjamini–Hochberg and (d) FDR vs FWER

(c) Thresholds $jq/m = j \cdot 0.10/6$:

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.007	0.009	0.020	0.040	0.320
jq/m	0.0167	0.0333	0.0500	0.0667	0.0833	0.1000
$p_{(j)} \leq jq/m?$	yes	yes	yes	yes	yes	no

Largest such j : $L = 5$ ($0.040 \leq 0.0833$; note $0.320 > 0.1000$) $\Rightarrow$ reject *all* of $H_{(1)}$ to $H_{(5)}$: **5 rejections** (nested: Bonferroni $2 \subseteq$ Holm $3 \subseteq$ BH 5). *Guarantee*: $\text{FDR} = E[V/\max(R,1)] \leq q = 0.10$ — *on average* at most 10% of rejected hypotheses are false discoveries (no bound on $\Pr(V \geq 1)$; that is FWER).

(d) Control the **FWER** (Bonferroni/Holm). This is a *confirmatory* study feeding a regulatory decision, so even a *single* false positive is costly (approving an ineffective drug); with only $m = 4$ endpoints the power lost to strict control is small, and the strong guarantee $\Pr(V \geq 1) \leq \alpha$ is exactly what regulators require.

Problem 7 — task

Problem 7 (20 P) — Cumulative essentials

- (a) [4 P] True or false: “with a flexible enough model and enough data the expected *test* MSE at a point can be driven to zero”. Justify.
- (b) [4 P] Logistic regression for passing an exam: $\hat{\beta}_1 = 0.8$ for hours studied ($e^{0.8} = 2.226$): effect of one more study hour on the *odds*; why is there no single number for the effect on the *probability*?
- (c) [4 P] One reason to prefer 5- or 10-fold CV over LOOCV and one over a single validation split.
- (d) [2 P] $p = 1,200$ SNPs; only about a dozen truly matter: ridge or lasso? Why?
- (e) [6 P] Match to {linear regression; lasso; logistic regression; random forest} — each exactly once: (i) earnings per extra school year with CI; (ii) customer lifetime value from 3,000 features ($n = 250$) with a short driver list; (iii) 30-day readmission yes/no with odds reported to administrators; (iv) bike-share ridership from hundreds of interacting variables; accuracy only.

Lecture slides: Chapters 2–6 (Lectures 1–8).

Problem 7 — solution (a) to (c)

Solution P7 — (a) to (c)

(a) False. Expected test $\text{MSE} = \text{Var}(\hat{f}) + [\text{Bias}(\hat{f})]^2 + \text{Var}(\varepsilon)$; the last term — the *irreducible error* $\text{Var}(\varepsilon) > 0$ — is noise no model can explain and a hard lower bound. Flexibility and data shrink the reducible part (bias, variance) towards 0, but the test MSE can never fall below $\text{Var}(\varepsilon)$.

(b) One more study hour adds 0.8 to the log-odds, i.e. *multiplies* the odds of passing by $e^{0.8} = 2.226$ (about +123%) — the same factor at every x . The *probability* change depends on the position on the S-curve: large near $p \approx 0.5$, tiny near 0 or 1 $\Rightarrow$ no single number.

(c) *vs. LOOCV*: the n leave-one-out fits are trained on nearly identical data $\Rightarrow$ their errors are highly correlated $\Rightarrow$ the LOOCV average has *higher variance* (and costs n fits). *vs. validation split*: a single split is highly variable and overestimates test error since only part of the data trains the model (*bias*); k -fold trains on $(k-1)/k$ of the data and averages k estimates $\Rightarrow$ less bias and more stability.

Problem 7 — solution (d) and (e)

Solution P7 — (d) and (e)

(d) Lasso: the ℓ_1 penalty sets coefficients *exactly to zero* $\Rightarrow$ variable selection; ideal for a sparse truth (about a dozen signals among 1,200 SNPs) and yields a small interpretable model. Ridge only shrinks and keeps all 1,200 predictors.

(e) Matching (each method once):

- (i) **Linear regression** — quantitative response, low dimension; goal is *inference*: effect size with confidence interval from OLS standard errors.
- (ii) **Lasso** — $p = 3,000 \gg n = 250$: OLS infeasible; ℓ_1 selects a short list of drivers.
- (iii) **Logistic regression** — binary response; coefficients give odds ratios $e^{\hat{\beta}_j}$: interpretable for administrators.
- (iv) **Random forest** — automatic non-linearities and interactions; strong accuracy with many heterogeneous predictors; interpretability not needed.